Bailin Liang

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EDUCATION

Jilin University, Changchun, China (Project 985 University) **Sept 2024–Jun 2027 (Expected)**
M.S. in Automotive Engineering (**Exam-Free Admission**), College of Automotive Engineering GPA: 88.21/100
Supervisor: Prof. Junnian Wang
Core Courses: Fundamentals of Modern Control Theory; Modern Numerical Computational Analysis and Methods; Applied Mathematic Statistics; Design and Control of New Energy Vehicles.
Awards include: WEICHAI Social Scholarship (**Top 1%**).

Jilin University, Changchun, China **Sept 2020–Jun 2024**
B.Eng. in Automotive Engineering, College of Automotive Engineering GPA: 87.89/100
Core Courses: Automotive Structure (Bilingual Teaching); Automotive Theory; Vehicle Design; The Basis of Control Engineering; Theoretical Mechanics; Calculus; Linear Algebra; Probability and Statistics.

RESEARCH EXPERIENCE

Nonlinear Shimmy Dynamics Analysis of Highly Integrated In-Wheel-Motor Driving Units *Apr. 2025–Present*
Sub-project of **Key Technologies for Highly Integrated In-Wheel-Motor Driving Units**
National Key R&D Program of China (Grant No. 2024YFB2505100)

- Developed a high-fidelity 13-DOF MATLAB/Simulink shimmy dynamics model for highly integrated in-wheel-motor driving units (HIDUs) with independent kingpin steering to investigate nonlinear shimmy mechanisms and parameter sensitivity. Built an Adams Car multibody model for comparative validation and implemented the workflow as an analysis software tool, laying the foundation for further research into generalized shimmy dynamics on multi-wheel platforms.
- Based on a simplified nonlinear shimmy model, proposed a nonlinear dynamics analysis framework for HIDUs, combining complex variable averaging (CX-A) and eigenspace-projected shooting for bifurcation analysis. Developed an FPSA method for efficient parameter evaluation and used energy-flow analysis to reveal parameter-dependent shimmy suppression mechanisms.

Coordinated Shimmy Suppression in Multi-Wheel Corner-Module Vehicles via Neural-Residual DiffTune-MPC *Sept. 2026–Present*

- **Generalized Shimmy Dynamics Modeling:** Developed a generalized nonlinear shimmy model for multi-wheel corner-module vehicles, capturing inertial coupling among wheel-shimmy motion, suspension roll, and wheel rotational oscillations, alongside lateral tire relaxation, steering backlash, and dry friction. Its modular architecture enables rapid adaptation across chassis configurations.
- **Nonlinear Shimmy Dynamics Analysis:** Investigating nonlinear shimmy dynamics of multi-wheel corner-module vehicles, focusing on how steering backlash, dry friction, and inter-subsystem dynamic coupling reshape stability boundaries and induce complex shimmy responses characterized by periodicity, quasiperiodicity, slow amplitude modulation, multistability, and nonzero-mean oscillations.
- **Shimmy-Aware Neural-Residual DiffTune-MPC:** Developing a hybrid prediction model that augments a physics-guided nominal model with a neural residual to capture unmodeled nonlinear dynamics and reduce prediction mismatch. A shimmy-aware DiffTune-MPC scheme tunes the inner MPC cost weights against a closed-loop outer objective that minimizes band-limited shimmy energy and promotes oscillation-envelope contraction, enabling coordinated drive–brake–steer control for shimmy suppression.

Generalized Fault-Tolerant Control for Multi-Wheel Corner-Module Vehicles via Reconfigurable MPC *Nov. 2025–Jul. 2026*

- Proposed a generalized fault-tolerant control framework integrating automatic topology identification, health-aware MPC reconfiguration, and physical latent-space optimization. The framework enables automatic adaptation to arbitrary multi-axle chassis configurations, maintaining stability and tracking accuracy under diverse faults while improving computational efficiency.
- Validated GFT-RMPC across 18 steering, driving, and composite fault scenarios, demonstrating superior stability and tracking performance over baseline methods for both 4-Wheel and 6-Wheel configurations. Hardware-in-the-loop tests further verify real-time execution within 12.36 ms, confirming its cross-configuration adaptability and practical deployability.

Architecture Design of Multi-Mode Power-Multiplexed Electric Wheel Corner Modules *Oct. 2025–May 2026*

- Proposed diverse novel power-multiplexing electric wheel corner-module architectures to reduce structural redundancy arising from function-stacking in conventional designs. By sharing a single power source through switchable transmission paths, these architectures support efficient multi-mode power-multiplexing for applications such as kingpin/omnidirectional dual-mode steering, land/air dual-mode mobility, and multi-mode driving–braking control.

Wheel-End Energy Autonomy for Distributed-Drive Electric Vehicles *Jan. 2024–Oct. 2024*

- Developed a wheel-end supercapacitor–battery architecture for distributed-drive electric vehicles, reducing energy-transfer losses and battery stress through localized energy management. Completed topology selection and parameter matching, ADVISOR/Simulink modeling, power-split control, and regenerative-braking optimization. Further integrated the architecture into corner modules with onboard energy storage, enabling local energy recovery, storage, and reuse while providing wheel-end energy autonomy and redundancy.

PUBLICATIONS

- **Bailin Liang**, Junnian Wang, Zidong Zhou, Guoshun Cai, and Yong Wang. "Generalized Fault-Tolerant Control for Multi-Wheel Corner-Module Vehicles via Reconfigurable MPC." *ISA Transactions* (JCR Q1, IF: 6.6; under review).
- **Bailin Liang**, Junnian Wang, Yitong Wang, Chunlin Zhang, Yinpu Qi, and Heping Ling. "Nonlinear Shimmy Dynamics Analysis of Highly Integrated In-Wheel Motor Driving Unit with Independent Kingpin Steering for Electric Vehicles." *Nonlinear Dynamics* (JCR Q1, IF: 5.7; under review).
- Junnian Wang, Yitong Wang, Chunlin Zhang, Zhiqin Li, Liyuan Liu, Wentao Hu, **Bailin Liang**, and Yinpu Qi. "A Review of Configuration Classification, Application Control and Future Trends of Highly Integrated Driving Corner Modules for Electric Vehicles." *Green Energy and Intelligent Transportation* (JCR Q1, IF: 21.5; under review).

PATENTS

As the **first student inventor**, I have authored **10 granted or published invention patents**, including **one published U.S. patent application**, and participated in **21 granted or published invention patents** in total.

- Kingpin Steering and Omni-Directional Steering Dual-Mode Power-Multiplexing Electric Wheel Angle Module. (**US20260200519A1**, published 16 Jul 2026; CN119749689B, granted 16 Sep 2025.)
- Design and Control Method of a Distributed Supercapacitor Energy Storage System for Electric Wheel Corner Modules. (CN119388982B, granted 10 Oct 2025.)
- Land-Air Power-Multiplexing Flying Car Using Electric Wheel Corner Modules. (CN121492543A, published 10 Feb 2026.)
- A Control Method for a Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287825A, published 11 Jul 2025.)
- Multi-Link-Braking Integrated Drive-Braking Electric Wheel System. (CN120481601A, published 15 Aug 2025.)
- Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287824A, published 11 Jul 2025.)
- Highly Integrated Driving Unit with Electronically Adjustable Kingpin Caster Angle. (CN120270338A, published 8 Jul 2025.)
- Electronic Wedge Braking System with a Sprung Actuator. (CN120288017A, published 11 Jul 2025.)
- Multi-Mode Redundant Integrated Drive-Braking Electric Wheel System. (CN120287823A, published 11 Jul 2025.)

INTERNSHIP

XPENG Motors, XNGP Team Guangzhou, China
Autonomous Driving R&D Intern *Jan 2024–May 2024*

- Studied XPENG's XNet-XBrain-XPlanner autonomous driving framework to understand XNGP's modular end-to-end pipeline.
- Contributed to the large-scale data closed loop for urban road testing, takeover events, corner cases, and OTA iterations by identifying failure and abnormal takeover scenarios and analyzing root causes.

Gspeed Formula Racing Team of Jilin University Changchun, China
Member of Suspension Group *Dec 2020–Dec 2023*

- Designed and modeled critical suspension components using CATIA and produced engineering drawings in AutoCAD.
- Developed the force analysis table for corner-wheel components by correlating hardpoints with vehicle performance targets.
- Participated in the full cycle of suspension system assembly, maintenance, and tuning, contributing to Gspeed's two consecutive **Formula Student Combustion China** (FSCC) **championships** in the 2021–2022 and 2022–2023 seasons.

AWARDS & HONORS

- WEICHAI Social Scholarship (**Top 1%**) (2025∼2026)
- Third-Class Academic Achievement Scholarship (2025∼2026)
- Graduate Academic Scholarship $\times\mathbf{3}$ (2024∼2027)
- Outstanding Graduate Student Scholarship $\times\mathbf{2}$ (2025∼2027)
- Outstanding Graduate of College (2023∼2024)
- Outstanding Student of College $\times\mathbf{2}$ (2021∼2023)
- Second-Class Undergraduate Scholarship $\times\mathbf{2}$ (2021∼2023)
- Third-Class Undergraduate Scholarship $\times\mathbf{2}$ (2020∼2021, 2023∼2024)
- Arts and Sports Scholarship $\times\mathbf{4}$ (2020∼2024)
- Jilin University Top 10 Class Award (collective honor) (2023∼2024)

SKILLS

Application & Coding: MATLAB/Simulink, Python, C/C++, CATIA, AutoCAD.
Languages: English (Professional); Mandarin Chinese (Native)